Reactjs vs Angularjs is like blurb against a book

Technology	React	AngularJS	Angular 2
Author	Facebook community	Google	Google
Type	Open source JS library	Fully-featured MVC framework	Fully-featured MVC framework
Toolchain	High	Low	High
Language	JSX	JavaScript, HTML	TypeScript
Learning Curve	Low	High	Medium
Packaging	Strong	Weak	Medium
Rendering	Server Side	Client Side	Server Side
App Architecture	None, combined with Flux	MVC	Component-based
Data Binding	Uni-directional	Bi-directional	Bi-directional
DOM	Virtual DOM	Regular DOM	Regular DOM
Latest Version	15.4.0 (November 2016)	1.6.0	2.2.0 (November 2016)

React vs Angular comparison: library against framework

Angular is one of the most popular javascript frameworks and like other similar software suits it offers multiple out-of-the-box solutions and designs. At a time when, React ecosystem comprises any number of composable, targeted online tools and ReactJS acts as one of the building blocks.

Advantages of angularjs

Global community support is one of the factors, that can easily make Angular the best javascript framework. Developers and designers constantly collaborate and contribute to the community, increasing credibility and reliability of the framework.

It is a **full-fledged framework** that can run in any browser or platform. Moreover, it is consistent, overwhelmed with ready-made tools, ng components are robust and quite mature, as contrasted with React.

Two-way data bind is probably the top feature, as it diffuses the impact after every minor data change and does way with the need for further effort with data sync in view and model.

Given the fact that our company makes active use of ng2, it is essential to include react vs angular 2 comparison as well.

TypeScript is an enhanced JS super-set that supplies optional static type checking, object-based programming patterns, and high-performance typing features.

Owing to **component-based architecture** components have deep binding and each of them comprises elements with only relevant functionality. What is more, they are loosely coupled and properly encapsulated. Such approach makes components easily reusable, enhance their testability and further maintainability.

Read also: The 5 Best Project Management Software: Does Your Company Need a Custom Solution?

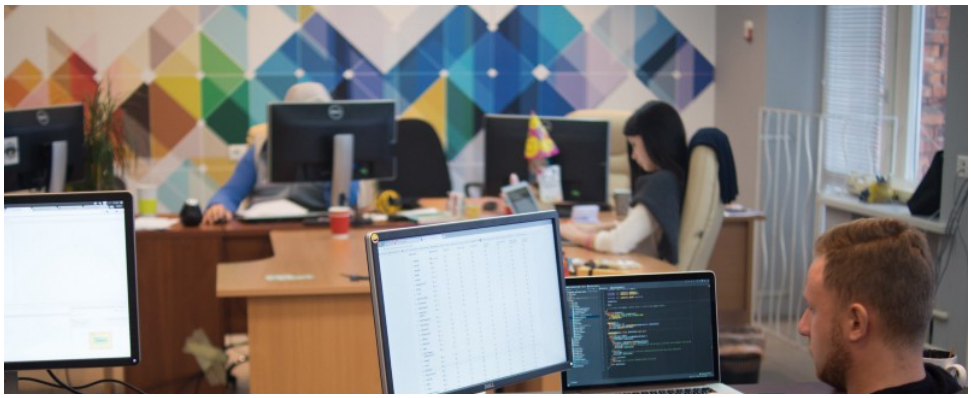
Advantages of reactjs

JSX is a JS syntax that enables HTML quotes and usage of HTML tag syntax for subcomponents rendering. It promotes building of machine-readable code and provides ability to compound components in one compile-time verified file.

Prompt rendering is among the best features of React that gives a significant edge over Angular. The technology comprises smart methods to mitigate the amount of DOM operations, optimize and accelerate the updates process. Virtual DOM (Document Object Model) is of great use while handling vast databases.

The core difference between reactjs and angularjs is that React is **JS-centric**, while ng2 remains HTML-centric. JavaScript is far more robust, than HTML, that makes React far more simple, focused and consistent.

React.js embeds HTML into JavaScript, while others continue putting JS into HTML.



Disadvantages of angularjs

Despite a comprehensive and clear manual, **steep learning curve** and **complexity** are named among the main weak points of Angular.js. Like any other client-side rendering technology in javascript framework comparison list, programmers should place special emphasis on security to make apps reliable and safe. Though, with introduction of **Angular Universal** and pre-rendering option in ng2 this issue was defused.

Disadvantages of react.js

Comparing react vs angular performance, first of all it's worth mentioning that reactJS is **not a full-scale framework** and for this very reason integration of the UI library into a common MVC framework requires deeper programming knowledge. It is still young and not mature, considering tutorial volumes, limited ecosystem, etc.

Apart from pros and cons of reactjs, we should also mention **Flux** that is frequently applied for adding a structure and architecture to react app. Usage of both technologies can become a challenge for non-experienced programmer, as it lacks structured and comprehensive documentation or guide.

Conclusion

React and Angular offer completely diverse approaches to web application development for startup, small and midmarket businesses. Both technologies are powerful and flexible, while none of them is worse or better, than the other. Depending upon custom app goals and particular system constraints, developers can run from ng2 to React, and back.

Opting for Angular, it usually assumes the creation of core skeleton for a front-end app, whereas React.js can be applied to improve its specific parts. Moreover, it can be integrated with other frameworks, like Backbone or even well-known Angular.

#The Meta Stuff

Attribute	AngularJS	Angular 2	React
Churn	Reduced	Reduced	High
Tooling	Low	High	High
Code Design	JS into HTML	JS into HTML	JavaScript Centric
JavaScript "Fatigue"	Less	Less	More

These tend to be high conversation areas that get started on Reddit and Hacker News.

#The Other Information

Attribute	AngularJS	Angular 2	React
DOM	Regular DOM	Regular DOM	Virtual DOM
Learning Curve	High	Medium	Low
Packaging	Weak	Medium	Strong
Abstraction	Weak	Strong	Strong
Debugging General	Good HTML / Bad JS	Good JS/Good HTML	Good JS / Bad HTML
Debug Line NO	No	No	Yes
Unclosed Tag Mentioned?	No	No	Yes
Fails When?	Runtime	Runtime	Compile-Time
Binding	2 Way	2 Way	Uni-Directional
Templating	In HTML	In TypeScript Files	In JSX Files
Component Model	Weak	Strong	Medium
Building Mobile?	Ionic Framework	Ionic Framework	React Native
MVC	Yes	Yes	View Layer Only
Rendering	Client Side	Server Side	Server Side

Most topics I have seen on Reddit, Hacker News, and other social media sites tend to only cover and talk about the first two tables. People get stuck on the thought of JavaScript Fatigue, JSX, TypeScript, Churn, and Size. I hope I have helped to break some of that by including all of the other information too which shows some holes in both frameworks.

I think some of the most interesting differences are the size, community, debugging, DOM, and mobile. When talking about mobile React has its own way of building mobile apps through React Native. Angular doesn't try to tackle it's own mobile and allows others like Ionic Framework to do this for it.

While the AngularJS 1 community is very large, the Angular 2 community is just gaining ground. This is in part due to Angular 2 still being in beta, but the traction it has already gotten is pretty outstanding. React's community continues to grow and become more diverse as well.

When looking at sizes, there are some distinct differences between React and Angular. While Angular 2 is currently at around 764k in size, they are going to be trimming a lot of that off in the coming months. React also does not do as much of the MVC model as Angular so it can get away with having a much smaller size. However if you add in the extras to React to make it fully MVC then you will still come out with a smaller framework size than Angular 2.

Debugging is sometimes the hardest thing to do. I think it is interesting how each framework handles debugging. Angular 2 and it's runtime debugging tends to give you less information than React and it's compile time debugging.

Finally, I wanted to talk a second about the Virtual DOM that React uses. This was a new concept to me, and I looked at it more in depth. This concept seems to be much faster since you don't have to work with the heavyweight parts that the real DOM has. Manipulating and changing a virtual copy of your DOM that is lightweight, and then only pushing the changes that are shown when doing a diff on the DOM is pretty awesome.

After spending a few days researching and reading other articles on these two frameworks, I have found that there is a lot of give and take when it comes to comparing the two. On some things Angular is better and on others, React is better. As always you should asses the project that you are working on to find out what best suits your needs. Maybe this comparison chart can help you in that decision.

How Easy Is It To Get Started?

JSX is optional, no workflow required!	Uses HTML & CSS in a well-known way	Uses HTML & CSS in a well-known way
Can be dropped into existing web apps/ views	Core features for SPAs are included and fit in nicely	Can be dropped into existing web apps/ view
JSX looks strange, working with JS-only maybe feels wrong	Learning TS and key features like DI can be intimidating	Intuitive syntax and core features provided by core team
For SPAs, you need to add a lot of key features manually	A complex project setup/ build workflow is required	Some key features for more complex apps have to be added manually

View (Need additional Flux/Redux to make the backend)	MVC	
Virtual DOM	DOM	
1 way bind	2 way bind	
ES6 JS, JSX / Mix of improper JS and HTML	TypeScript / Proper HTML	Any JS-type language
	TypeScript need offline compilation	
Easy creation of functional stateless components, good for composition	Focus on app	Components
Need many community extensions (eg router, form validator …)	All extensions, modules and dependencies within Angular core	Provide some dependencies in core, eg have router but no form validator
1 page and multi-page	1 page apps only	1 page and multi-page
	Modularity, Scalability	
Startups	Enterprise	
FB licenses	Poor backward compatibility	
	Best CLI	Best Documentation
Multi Device (Better than Angular's Cordova		